MIL-F-80056B
25 January 1979
SUPERSEDING
MIL-F-80056A
22 April 1970

MILITARY SPECIFICATION

FURNACES, HEAT TREATING, ELECTRIC, BATH TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers bath type, electrically heated, heat treating furnaces of the round and rectangular pot design.

1.2 Classification.

1.2.1 Types and classes. The furnaces shall be of the following types and classes, as specified (see 6.2.1).

Type I - Round Pot, External Resistance Heated
  Class A - 600° F to 1650°F operating temperature
Type II- Rectangular Pot, Electrode Heated
  Class A - 600°F to 1200°F operating temperature
  Class B - 1200°F to 1700°F operating temperature
  Class C - 1700°F to 2400°F operating temperature
Type III - Rectangular Pot, Immersion Resistance Heated
  Class A - 350°F to 750°F operating temperature

1.2.2 Sizes. The size of the furnace shall be as specified (see 3.6 and 6.2.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3424

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2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

STANDARDS

MILITARY

MIL-STD-461 Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

* AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C96.1 - Temperature Measurement Thermocouples
Z2101.1 - Metric Practice (ASTM-E-380)

(Applications for copies should be addressed to the American National Standards Institute Dept. 969, 1430 Broadway, New York, NY 10018.)

* OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF 1970

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and Amendments	Occupational Safety and Health Administration Standards

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA) STANDARDS

ICS - Industrial Controls and Systems

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2011 L St. N.W. Wash, DC 20037.)

JOINT INDUSTRIAL COUNCIL STANDARDS (JIC)

EGP-1 Electrical Standards for General Purpose Machine Tools

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, VA 22101.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete furnace for first article inspection and approval (see 4.2 and 6.4).

3.2 Design. The furnace shall be of the bath type, electrically heated and of the manufacturer current design. The furnaces shall be new and the design shall be in accordance with 3.2.1, 3.2.2 and 3.2.3 for the type specified.

3.2.1 Type I. The Type I furnace shall be of the round pot design, externally resistance heated. Heating elements shall surround the pot and shall be located on the insulation wall in the space between the pot wall and insulation.

* 3.2.2 Type II. The Type II furnace shall be of the rectangular pot design, internally heated by either suspended immersion-type electrodes, or internally heated by built-in submerged electrodes.

3.2.3 Type III. The Type III furnaces shall be of the rectangular pot design internally heated by immersion resistance elements. When required, the Type III furnace shall be provided with quench pumps, heat exchangers, and agitation devices for quenching processes. Quench equipment and performance requirements shall be as specified (see 6.2.1).

* 3.2.4 Safety and health requirements. The furnace shall be provided with safety features and devices for the protection of personnel, equipment and the work being processed. The safety system of the furnace shall conform with the standards promulgated under OSHA applicable to the furnace itself. Additional requirements for safety and health shall be as specified (see 6.2.1 and 6.5).

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* 3.2.5 Lubrication. When necessary to reduce wear and maintain efficiency, means shall be provided for lubrication of all sliding and rotating parts, except non-additive pack-seal bearings. Lubrication systems shall be designed and built to prevent any contamination of bath, components, material or equipment.

3.2.6 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modifications of parts or furnace.

3.2.7 Accessibility. All parts subject to wear, breakage, or distortion and all parts which require periodic lubrication or maintenance shall be accessible for adjustment, replacement, and lubrication as applicable.

3.2.8 Material. All material used in construction of the furnace shall be of uniform quality, and shall conform in composition, heat treatment, and suitability to produce a furnace to meet the requirements described herein.

* 3.2.9 Reclaimed materials. The furnace may contain reclaimed materials provided such materials will not jeopardize its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which restores them to the same chemical composition and physical properties as the materials originally selected for use on the furnace.

* 3.2.10 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the furnace. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in the U.S. Customary System of Units (US). These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and methods specified in ANSI Z210.1.

3.3 Construction. The furnace shall be constructed so that when installed and connected to the specified power source, it can be used for any purpose described herein. All parts and necessory equipment shall be functionally suitable for the purpose intended. Repaired parts shall not be used.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. In no event shall such processes as welding, peening, plugging, or filling with solders or pastes be used on castings or forgings to reclaim any defective parts for use on the furnaces.

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3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where these operations are required in the original design. These operations shall not be employed as repair measures for defective parts.

3.3.3 Fastening devices. All screws, pins, bolts, and similar fasteners shall be installed with means for preventing loss of tightness. Such fasteners subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled unless sharpness is required to perform a necessary function.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the furnace shall be painted in accordance with the manufacturer's standard commercial practice.

3.3.6 Threads. All threaded parts shall conform to FED-STD-H28.

* 3.3.7 Dials. Dial diameters shall be such that the graduations may be easily read from the normal operating position. Dials shall be permanently and legibly engraved, etched or marked on a non-glare background.

* 3.3.8 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.4 Electrical requirements. The furnace shall be provided with a magnetic contactor, a separate main line disconnect switch with provisions for locking in the "off" position, and all necessary electrical heating elements, electrodes, controls relays, transformers, ammeters, signal lights and switches, prewired where possible for operation upon connection to source of power. All electrical equipment and wiring shall be in accordance with NEMA-ICS Standards.

3.4.1 Power supply. Unless otherwise specified (see 6.2.1), furnaces shall operate on a nominal 230 volt, 60 Hz, power supply. Furnaces up to and including 5 KW input rating shall operate on single phase. Furnaces

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above 5 KW inputs, up to 40 KW input rating inclusive, shall be either single phase or 3 phase. Furnaces above 40 KW input rating shall operate on 3 phase. All power actuated, manually adjusted and automatic controls shall operate on and shall be wired for 115 volt, 60 Hz current.

3.4.2 Electrical connections. When necessary to maintain efficiency and to protect the equipment from excess heat, the terminals connecting the transformer to the electrodes shall be water cooled. Terminals of the electrical heating elements and electrodes shall have at least twice the current carrying capacity of the heating elements and electrodes, and shall be of the solderless type. The electrodes shall be brazed to steel sections with copper connections. Other electrical connections and terminal shall be of the pressure or soldered type. When soldered connections are used, the conductors and terminal shall be mechanically secured before soldering. Soldering flux, when used, shall be of a resin base.

3.4.3 Electromagnetic interference control. When specified (see 6.2.1), furnaces shall have electromagnetic interference control in accordance with the applicable requirements of MIL-STD-461.

3.5 Performance requirements. The furnace shall be capable of continuous operation as its maximum operating temperature. Temperature variation within the working area of the bath shall be not more than $\pm 5^{0}$ F from any preset temperature within the operating range of the furnace, after the temperature has stabilized at set point. The furnace shall be such design and construction that it will provide a stable support for the bath material and a load in the bath of not less than 125 pounds per cubic foot of bath capacity. Unless otherwise specified (see 6.2.1), minimum power input requirements shall be as specified in the tables herein for the type, class, and size furnace required.

3.6 Size. The size of the furnace should be selected from the sizes listed in table I through V for the type and class required. The dimensions specified shall be the clear inside minimum working dimensions of the bath. If a size different from the sizes listed in the tables herein is required, the minimum power input requirements shall be as specified (see 6.2.1).

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TABLE I. Round Pot, External Resistance Heated

Type I - Class A

Size Furnace Bath Working Dimensions in Inches		Input KW Not Less Than
Diameter	Depth	
8	10	8
10	14	17
12	18	20
16	18	27
16	30	34
24	24	40

TABLE II. Rectangular Pot, Electrode Heated

Type II - Class A

Size Furnace Bath Working Dimensions in Inches			Input KW Not Less Than
Length	Width	Depth	
12	7	22	15
15	9	22	18
18	15	22	25
24	15	22	32
36	20	24	50

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TABLE III. Rectangular Pot, Electrode Heated
Type II - Class B

Size Furnace Bath Working Dimensions in Inches			Input KW Not Less Than
Length	Width	Depth	
9	5	22	22
9	7	30	28
12	11	22	40
15	9	30	40
24	15	22	70
24	15	30	80

TABLE IV. Rectangular Pot, Electrode Heated
Type II - Class C

Size Furnace Bath Working Dimensions in Inches			Input KW Not Less Than
Length	Width	Depth	
9	7	22	35
12	9	22	50
15	9	22	60
18	12	22	90
21	15	30	90
24	15	30	100

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TABLE V. Rectangular Pot, Immersion Electrode Heated

Type III - Class A

Size Furnace Bath Working Dimensions in Inches			Input KW Not Less Than
Length	Width	Depth	
12	12	24	10
18	12	24	10
24	17	24	20
24	17	30	20
36	17	24	20
36	17	30	20
48	17	24	30
48	17	30	30

3.7 Components and systems.

* 3.7.1 Shell. The outer shell of each furnace (except Type II, submerged electrode design) shall be fabricated of sheet steel, reinforced and welded on all sides and the bottom. The shell thickness shall be not less than 10 gage for the Type I furnace and not less than 3/16 of an inch for Type II immersion electrode design and Type III furnaces. The submerged electrode design furnace may have a ceramic outer shell. The top of the bath (excluding cover) shall be between 32 and 38 inches above floor level, except for furnaces requiring greater height for reasons of capacity. Unless otherwise specified (see 6.2.1), the Type II and Type III furnaces shall be mounted on steel I-beam skids or similar material to elevate the bottom of the furnace not less than 3 inches above the floor. The top of the cabinet shall be constructed of material to prevent distortion. Cabinet construction shall prevent contact between insulation and bath material.

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3.7.2 Drain. The Type I furnace shall be provided with drainage facilities for the bath material in case of pot failure. The insulation shall be sloped to the drain.

3.7.3 Cover.

3.7.3.1 Type I. When specified (see 6.2.1), the Type I furnace shall be equipped with an insulated swivel mounted, split type cover.

3.7.3.2 Type II and Type III. When specified (see 6.2.1), the Type II and Type III furnace shall be provided with a hand operated one or two piece rolling cover. The cover shall be insulated or made of castable refractory. When specified (see 6.2.1), the cover shall be electric motor powered.

* 3.7.4 Insulation. Each furnace shall be insulated with insulating fire brick, block, ceramic fiber, or pre-formed insulation selected to insure long life and minimum heat loss. Powdered, shredded, or granular insulation is not acceptable. Quality and thickness of the insulation shall be such that the outside surface temperature of the furnace will not exceed the maximum surface temperature specified in 3.7.4.1, 3.7.4.2, and 3.7.4.3 for the respective type and class furnace specified.

3.7.4.1 Type I, class A. The insulation adjacent to the heating elements shall have a temperature rating of not less than 2000°F and the temperature rating of the back-up insulation shall be not less than 1600°F. The maximum outside surface temperature of the furnace shall not exceed 80°F above ambient air temperature with the bath temperature stabilized at 1600°F.

3.7.4.2 Type II, class A and Type III, class A. The insulation adjacent to the steel pot shall have a temperature rating of not less than 400°F above the maximum temperature of the furnace. The maximum outside surface temperature of each furnace shall not exceed 60°F above ambient air temperature with the bath of the Type II furnace stabilized at 1100°F and the bath of the Type III furnace stabilized at 750°F.

3.7.4.3 Type II, class B and C. The Type II, class B and C furnaces, when furnished with ceramic pots, shall have back-up insulation consisting of refractory lumnite cement block or pre-formed insulation of sufficient quality and thickness to insure not more than 8 percent by volumn bath material (salt) absorption during the life of the furnace, or a secondary leakproof steel pot separating the ceramic pot from the insulation. The secondary steel pot, if used, shall be not less than 3/16 of an inch thick and welded inside and out. Maximum outside surface temperature for the Type II, class B furnace shall not exceed 80°F above ambient air tempera-

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ture with the bath temperature stabilized at 1600°F, and 140°F above ambient air temperature for the Type II, class C furnace, with the bath temperature stabilized at 2200°F.

3.7.5 Pot. Each furnace pot shall be capable of continuous operation at the maximum operating temperature specified in 1.2 for each type and class furnace without sagging or warping. Metal pots shall be not less than 3/8 inch thick and shall be free from porosity. Additional characteristics shall be in accordance with 3.7.5.1, 3.7.5.2, and 3.7.5.3 for the type and class of furnace specified.

3.7.5.1 Type I. The Type I pot shall be cylindrical with a rounded bottom and shall be cast from an alloy containing not less than 35 percent nickel and 18 percent chromium. Means shall be provided for properly sealing the space between the pot flange and top plate.

3.7.5.2 Type II and Type III, class A. Unless otherwise specified (see 6.2.1), the pot for the Type II and Type III, class A furnace shall be fabricated from aluminized low-carbon steel plate. All seams shall be welded inside and out.

3.7.5.3 Type II, class B and C. Unless otherwise specified (see 6.2.1), the pot for the Type II, class B and C furnace shall be fabricated from pre-formed and pre-fired salt-resistance ceramic material.

3.7.6 Heating elements.

3.7.6.1 Type I. Heating elements shall be of the rod or ribbon type, fabricated form an alloy of 80 percent nickel and 20 percent chromium. The elements shall be mounted on the heating chamber wall in the space between the pot wall and insulation, properly insulated from each other, and distributed in such a way to produce uniform temperature specified in 3.5. Heating elements shall not be located in the bottom of the heating chamber. Power density on the heating elements shall not exceed 15 watts per square inch of radiation surface area.

* 3.7.6.2 Type II. The II furnace shall be heated either by suspended immersion type electrodes or shall be heated by built-in submerged electrodes. The electrodes shall be spaced and located in the bath in such a way that full transformer power may be obtained through the transformer taps. The portion of the electrode which is subject to salt corrosion shall be stainless steel.

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3.7.6.3 Type III. The Type III furnace shall be heated by sheath immersion resistance type elements. The elements shall enter the bath from the top and be provided with necessary protection to prevent being damaged by the work load.

* 3.7.7 Transformer. Unless otherwise specified (see 6.2.1), Type II furnaces designed to operate on single phase power shall be provided with a single phase, 7 tap, transformer. Furnaces designed to operate on 3 phase power shall be provided with a 3 phase, Scott connected, 7 tap transformer. Transformer cooling shall be either convection air or fan cooled when used with furnaces having an input rating up to 40 KW. Unless otherwise specified, furnaces with a rating above 40 KW shall be provided with a fan cooled transformer. When specified (see 6.2.1), transformers rated above 40 KW shall be water cooled. Transformer cooling shall be adequate to prevent a temperature rise in excess of its permitted maximum designed safe operating temperature. Each transformer shall be provided with overload and under-voltage protection.

3.7.8 Starting method. When required, an electric starting unit or starting torch kit, as specified (see 6.2.1), shall be provided for starting a frozen or new bath in the Type II furnace.

3.7.8.1 Starting torch kit. The kit shall consist of a burner, handle, extension tube, pressure gage, pressure regulator, and not less than ten feet of hose with suitable connections for connecting to commercial gas bottles.

3.7.8.2 Electric starting unit. The electric starting unit shall consist of an electric resistance type heating unit with transformer leads.

* 3.7.9 Thermocouples, sensors, and target tubes. Thermocouples shall conform to ANSI C96.1 and each thermocouple and sensor shall be provided with not less than 50 feet of suitable lead wire.

3.7.9.1 Type I, class A and Type II, class A and class B. The Type I, class A and Type II, class A and class B furnace shall be provided with two chromel-alumel type thermocouple assemblies for each furnace. The Type I furnace shall provide for locating one thermocouple in the heating chamber between the insulation and pot wall for the temperature controller and one thermocouple located in the bath for the excess temperature controller. The Type II, class A and class B furnace shall provide for locating two thermocouples in the bath, one for the excess temperature control and one for the temperature controller.

3.7.9.2 Type II, Class C. The Type II, class C furnace shall be provided with two radiation-type sensors and target tubes for each furnace, one for the temperature controller, and one for the excess temperature control or high-low controller.

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3.7.9.3 Type III. The Type III furnace shall be provided with one iron-constantan type thermocouple assembly for locating in the bath and connecting to the temperature controller.

* 3.7.10 Instruments. The instruments shall be of solid state design, shall be marked to indicate the particular type thermocouple for which it is calibrated and have automatic reference junction compensation in all instruments using thermocouples as the termperature sensing element. The controllers shall provide thermocouple upscale failsafe protection that will shut off the heat source at approximately 100°F above maximum temperature.

* 3.7.10.1 Type I and Type II. The Type I and Type II furnace shall be provided with a single indicating-recording, controlling type instrument. Unless otherwise specified, the instrument shall be of the time proportioning type control and shall operate a contactor through an intermediate relay system. When specified (see 6.2.1), the Type II, class C furnace shall be provided with saturable core reactor or a continuous proportional SCR control system. Either system furnished shall have an accuracy of 0.25 percent or better, of full scale reading and shall be capable of maintaining the set control temperature within ±2°F, within operating range of the furnace.

* 3.7.10.1.1 Control indicator scale. The temperature control indicator scale shall be approximately the same length as the temperature recorder chart width. The calibrated scale range shall be not less than 100°F above the maximum temperature of the furnace. The increments between successive temperature marks shall be not greater than 20°F, with the spacing of numbered marks not greater than 400°F. The control indicator shall be provided with two pointers. One pointer shall be adjustable to the desired control temperature, and one pointer shall indicate the actual furnace temperature.

* 3.7.10.1.2 Temperature recorder. The temperature recorder shall be of the strip chart, self-balancing type. Unless otherwise specified, the calibrated chart width shall be not less than six inches, and the chart length shall be not less than 80 feet to the roll. The chart speed shall be one inch per hour. The chart mechanism shall be furnished in a dust-proof housing with a glass door that permits full view of chart width, and not less than seven inches of chart length. When required, the chart width shall be not less than 9 inches and provisions provided for changing chart speed to 4 and 6 inches per hour as specified (see 6.2.1).

* 3.7.10.1.3 Excess temperature controller. When specified (see 6.2.1), each furnace shall be provided with an excess temperature controller. The excess temperature control circuit shall be independent of the control circuitry. The controller shall provide for selecting the over-temperature

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control point and shall function to shut-off the heat source to the furnace when the over-temperature control point is reached. The excess temperature controller shall provide for furnace shut-down in the event of thermocouple failure, or failure of other components in the circuit. In case the furnace is shut-down by the excess temperature control system, it shall sound an audible alarm and the furnace shall remain off until started manually. Calibrated accuracy shall be within one percent of full scale reading.

* 3.7.10.1.4 High-Low temperature controller. When specified (see 6.2.1), in lieu of the excess temperature controller specified in 3.7.10.1.3, the furnace shall be provided with a Hi-Lo temperature controller. The Hi-Lo control circuit shall be independent of the control circuitry and the controller shall provide for selecting the high temperature control point and the low temperature control point. The Hi-Lo controller shall interrupt the heat source to the furnace when the Hi temperature control point is reached and sound an alarm. The temperature shall coast down to the low control point and the temperature shall be maintained at the low point and the alarm shall continue to sound.

* 3.7.10.2 Type III. Unless otherwise specified, the Type III furnace shall be provided with an indicating, non-recording, automatic temperature controlling system capable of maintaining the bath temperature within plus or minus 10^{0}F of control temperature. The control indicator shall have a calibrated scale length of not less than six inches, shall be calibrated in degrees Fahrenheit and in increments of 5^{0}F with the spacing of numbered marks not greater than 100^{0}F. The control indicator shall have a maximum range of not less than 100^{0}F above the maximum range of the furnace. When specified (see 6.2.1) the Type III, furnace shall be provided with the controller recorder specified for the Type I and II furnace and the instrument panel specified for the Type I furnace.

3.7.11 Instrument panel.

3.7.11.1 Type I. Unless otherwise specified (see 6.2.1), the Type I furnace shall be provided with one factory-wired, front opening floor-mounted, instrument panel conforming to JIC EGP-1 for mounting all instruments and accessory controls. All controls shall be suitably mounted within the panel. A separate enclosure shall be provided for mounting the power contactor individual and separate from the instrument panel. The panel shall provide on-off signal lights and flush mounting of all instruments.

3.7.11.2 Type II. Unless otherwise specified (see 6.2.1), the Type II furnace shall be provided with an instrument panel as specified for the

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Type I furnace. In addition to the panel, the Type II furnace shall be provide with a floor-mounted switch gear control cubicle totally enclosed and factory-wired with the switch gear components suitably mounted within the cubicle. The control cubicle shall conform to JIC EGP-1.

3.8 Optional equipment, material and parts. When required (see 6.2.1), the following optional equipment, material, and parts shall be furnished as specified.

a. Bath salt, quantity, and type shall be as specified.
b. Quench basket and fixtures. Design shall be as specified.
c. Special tools such as ladles, skimmers, electrode scrapers, etc.
d. Quench and wash tanks. Type and design shall be as specified.
e. Portable air operated pump for emptying molten baths.
f. Ventilating hood. Design and dimensions shall be as specified.
g. Repair parts such as extra electrodes, salt pot, chart paper, etc.

* 3.9 Nameplate. Unless otherwise specified, (see 6.2.1), a corrosion resistant metal nameplate, shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix keyed to the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input characteristics and ratings
Maximum operating temperature (degrees F)
Bath working dimensions (length x width x depth)
Date of manufacture
Contract or order number
Plant Equipment Code or National Stock Number
U S

* 3.10 Technical data. Technical data shall be furnished as specified (see 6.3).

3.11 Workmanship. Workmanship of the furnace and accessories shall be commensurate with the requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS.

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When first article approval is required under 3.1, first article inspection shall be performed. Unless otherwise specified (see 6.2.1), first article inspection shall comprise the examination in 4.5 and all tests in 4.6. Failure of the item to pass any examination or test shall be cause for rejection.

4.4 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.5, the tests in 4.6.1 and 4.6.2 and the inspection in 4.7. Failure of the item to pass any examination, test, or inspection shall be cause for rejection.

* 4.5 Examination. The furnace and equipment shall be examined for design, dimensions, construction, material, components, electrical and safety compliance, and workmanship to determine compliance with the requirements of this specification.

* 4.6 Tests. All test shall be conducted at an ambient temperature of $77^{\circ}F \pm 18^{\circ}F$. Test instruments and sensing elements used to perform the test specified for the furnace and instruments should have been checked against a standard potentiometer type instrument and calibrated thermocouple of known accuracy within the previous three months to read accurate within $\pm 2^{\circ}F$.

* 4.6.1 Instrument test. Indicating and recording, instruments shall be tested by impressing various predetermined voltages, simulating thermocouple outputs for different furnace temperatures to terminals where

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normally the thermocouples would be connected. These tests shall be made in accordance with and results checked against, standards established in ANSI C96.1. The instruments shall be tested through their operating range.

4.6.2 Circuit test. Each circuit of the electrical system shall be tested for dielectric strength, continuity, insulation-resistance and faulty grounds in accordance with the requirements of NEMA ICS.

* 4.6.3 Electromagnetic interference control testing. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.4.3 in accordance with Table II of MIL-STD-461.

4.6.4 Performance test. The heat treating furnace selected for test shall be filled to 2/3 pot capacity with appropriate bath material. An indicating or recording wattmeter shall be connected in the heater power circuit to indicate the power input. With all circuits energized, the power input shall be measured at the point of maximum power consumption between start and operating temperature and shall be not less than the minimum KW input specified for the type, class, and size furnace under test. The furnace shall then be operated at the temperature specified in 3.7.4 until equilibrium has been established. The outer shell temperature shall not exceed the maximum surface temperature specified for the type and class furnace under test.

4.6.5 Temperature variation. Thermocouples with suitable indicating instruments shall be so located in the bath as to determine that the temperature variation specified in 3.5 has not been exceeded. This test shall be conducted at both minimum and maximum furnace operating temperature.

* 4.7 Inspection of packaging. Packaging shall be inspected to determine compliance to the requirements of Section 5.

5. PACKAGING

* 5.1 Preservation, packing and marking. Preservation, packing, and marking shall be in accordance with MIL-F-3296. Level of preservation, and packing shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. Type I, Class A furnaces are intended primarily to be used in lead bath operations and operations requiring salts which

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are not compatible with ceramic materials. The Type II, Class A and Type III, Class A furnace may be used for most salt bath operations in the lower temperature range. The Type II, Class B and C furnaces are intended to be used in operations requiring neutral salts in the middle and upper temperature ranges.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.
b. Type and class furnace required (see 1.2).
c. Size furnace required, length, width, and depth, (Type I, diameter and depth) (see 1.2.2 and 3.6).
d. When first article is required (see 3.1 and 6.3).
e. Quench equipment and performance characteristics, when required (Type III only) (see 3.2.3).
f. Additional safety and health requirements when required (see 3.2.4).
g. If furnace is required to be configured in a specific measurement system (US or SI) state required system (see 3.2.10).
h. Painting, if different (see 3.3.5).
i. Operating voltage and phase, if different (see 3.4.1).
j. Electromagnetic interference control, when required (see 3.4.3).
k. Power input requirements (KW), if different (see 3.5).
l. Power input requirements, if the size required is different from the sizes listed in the tables (see 3.6).
m. Mounting skids (Type II and III only, if not required (see 3.7.1).
n. Cover (Type I) if required (see 3.7.3.1).
o. Cover (Type II) and Type III) if required, (specify manual or power operated) (see 3.7.3.2).
p. Pot material, if different (Type II and III, Class A) (see 3.7.5.2).
q. Pot material, if different (Type II, Class B and C) (see 3.7.5.3).
r. Transformer, if different (Type II) (see 3.7.7).
s. Water cooled transformer, if required (see 3.7.7).
t. Starting device, if required (Type II) (see 3.7.8).
u. Staturable core reactor or SCR controller, if required (see 3.7.10).
v. Chart width, if different (see 3.7.10.1.2).
w. Excess temperature controller, if required (see 3.7.10.1.3).
x. High-Low controller in lieu of excess temperature control, if required (see 3.7.10.4).

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y. Temperature control system, if different (Type III) (see 3.7.10.2), (specify if the same instruments and panel are required as specified for Type I and II furnace).
z. Instrument panel, if different (Type I) (see 3.7.11.1).
aa. Instrument panel and switch gear cubicle, if different (Type II) (see 3.7.11.2).
bb. Optional equipment, each item as specified (see 3.8).
cc. Nameplate, if different (see 3.9).
dd. First article inspection, if different (see 4.3).
ee. Quality conformance inspection, if different (see 4.4).
ff. Level of preservation, packaging, and packing required (see 5.1).

* 6.3 Contract data requirements. Required technical data such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

* 6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. A first article comprise a preproduction item or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments, regarding arrangement for examination, tests and approval of the first article.

* 6.5 Safety and health determination. In order that equipment integrated into the user's operational environment will comply with the standards promulgated under the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable; specific requirements concerning guarding the point(s) of operation, and other safety and health requirements should be specified.

* 6.6 Mercury prohibited. When the use of mercury or its compounds in the furnace and its components and accessories is to be prohibited, procurement documents should include a statement to that effect.

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6.7 Marginal notations. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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